

Haotian Song

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Education

National University of Singapore, Centre for Quantum Technologies

Ph.D. Candidate in Quantum Simulation

Singapore

Jan 2022 - Jan 2027 (Expected)

Dissertation: Quantum Simulation of Fermions in Disordered 2D Optical Lattices

Advisor: Prof. Kai Dieckmann

Xi'an Jiaotong University

B.S. in Physics (Honors)

Xi'an, China

Sep 2017 - Jun 2021

Tsien Hsue-shen Talented Program (top 10%); GPA: 89.37/100

Additional academic experience: University of Manchester, Texas A&M University, and Peking University School of Physics Summer School.

Research Experience

- **Quantum simulation of fermions in disordered 2D optical lattices.**
Built and operated an experimental platform for ultracold 6Li Fermi gases in disordered two-dimensional optical lattices, targeting transport, relaxation, localization, and cross-dimensional thermalization in controlled quantum systems.
- **2D optical lattice creation and characterization.**
Created far-detuned 1064 nm optical lattices, verified lattice loading and band structure through band mapping, and calibrated lattice depths using molecular BEC diffraction and amplitude modulation spectroscopy.
- **Accordion-lattice 2D confinement.**
Developed and operated a 532 nm vertical optical accordion lattice to compress a 3D gas into a 2D pancake geometry; loaded about 2×10^3 atoms with 2D density around $1.87 \times 10^8 \text{ cm}^{-2}$ and filling fraction per spin state around 0.26.
- **Speckle disorder engineering.**
Generated optical speckle disorder with a 532 nm beam and characterized speckle size through ex-situ imaging, autocovariance fitting, theory comparison, and numerical simulation.
- **Experimental control and data analysis.**
Used magnetic-field tuning to control interactions and optical power to control disorder strength; developed automation, absorption-image analysis, numerical fitting, and diffusion/localization simulation workflows.

Skills

- Experimental techniques: ultracold atom experiments, 6Li Fermi gases, molecular BEC preparation, Feshbach resonance control, absorption imaging, band mapping, amplitude modulation spectroscopy.
- Optics and laser systems: 1064 nm optical lattices, 532 nm accordion lattice, speckle disorder projection, optical alignment, laser locking and stabilization, PDH locking, AOM/EOM control.
- Laboratory systems: UHV systems, optical tables, magnetic-field control, laser diagnostics, experimental troubleshooting, integrated cold-atom platforms.
- Software and analysis: MATLAB, Python, Mathematica, Fortran, C++, numerical fitting, Monte Carlo simulation, image analysis, experimental automation, LaTeX.
- Research areas: quantum simulation, quantum transport, Anderson localization, cold-atom many-body physics, computational ghost imaging, numerical astrophysics.

Publications

- [Manuscript in preparation] Quantum Simulation of Fermions in Disordered 2D Optical Lattices.
- X. Nie, H. Song, W. Ren, Z. Zhang, T. Peng, M. O. Scully. 2024. Deep correlated speckles: suppressing correlation fluctuation and optical diffraction. *Photonics Research* 12(4), 804-812.
- H. Song, X. Nie, H. Su, H. Chen, Y. Zhou, X. Zhao, T. Peng, M. O. Scully. 2022. 0.8% Nyquist computational ghost imaging via non-experimental deep learning. *Optics Communications* 520, 128450.
- Z. Zuo, H. Song, H. Xue. 2021. Population synthesis on ultra-luminous X-ray sources with an accreting neutron star: Wind Roche-lobe overflow cases. *A&A* 649, L2.
- H. Song. Ultra-luminous X-Ray sources with wind Roche lobe overflow in Ring galaxies. Undergraduate thesis, Xi'an Jiaotong University.

Conference Presentations

- Progress Toward a Quantum Simulation of Fermions in Disordered 2D Optical Lattices, Rishav Koirala, Haotian Song, Athira Sreedevi, Kai Dieckmann, Institute of Physics Singapore - IPS Meeting 2024, Oct 02 - Oct 03, 2024, Nanyang Technological University, Singapore, contributed poster.
- Quantum Simulation with 2D Ultracold Fermions, Haotian Song, The 29th International Conference on Atomic Physics, Jun 14-19, 2026, Wuhan, China, poster.

Awards & Honors

- Outstanding Graduate Thesis Award (Top 1%), Xi'an Jiaotong University, 2021.
- Everest Scholarship, Xi'an Jiaotong University, 2021 and 2018.
- Tsien Hsue-shen Academic Research Award, 2021; Academic Research Award, Xi'an Jiaotong University, 2020.
- First Prize, 5th Chinese Undergraduate Physics Experiment Competition, 2019.
- Provincial Award, China Undergraduates Innovation and Entrepreneurship Competition, 2020.
- First Prize, Contemporary Undergraduate Mathematical Contest in Modeling, Shaanxi Province, 2018.

Additional Projects Experience

- **Ultra-luminous X-ray source population synthesis.**
Contributed to the A&A publication on accreting neutron-star ULXs with wind Roche-lobe overflow; implemented and analyzed numerical workflows for binary-evolution and population-synthesis modeling using Fortran and Python.
- **Computational ghost imaging and deep correlated speckles.**
Developed CNN-based computational ghost-imaging workflows for low-sampling reconstruction, including the published 0.8% Nyquist ghost-imaging work; contributed to deep-correlated-speckle research for improved computational imaging.

References

References available upon request; primary supervisor: Prof. Kai Dieckmann, Centre for Quantum Technologies, National University of Singapore.